MILITARY SPECIFICATION

INDICATOR, TACHOMETER, TRIPLE POINTER, HELICOPTER

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers design and all performance requirements for helicopter triple pointer tachometer indicators.

1.2 Classification. The indicators shall be of the following types:

	100 percent RPM Indicated		
	Engine 1	Engine 2	Engine 3
MS18099-1 Unlighted	4200	4200	4200
MS18099-2 Lighted	4200	4200	4200
MS18099-3 Unlighted	3794	3794	3900
MS18099-4 Unlighted	3900	3900	2640

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-Z-325	Zinc Plating, Electrodeposited, Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, ESSD, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6620

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SPECIFICATIONS (Continued)

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-G-9398	Generator, Tachometer, Two-Pole, High Temperature, Aircraft, Type MB-6
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-C-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature Quick Disconnect
MIL-G-26611	Generator, Tachometer GEU-7-A, Miniature
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover
MIL-B-81793	Bearing, Ball, Precision, for Instruments and Rotating Components

STANDARDS

Federal

FED-STD-595	Colors

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STANDARDS (Continued)

Military

MIL-STD-130	Identification Marking of U.S Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-756	Reliability Prediction
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of
MIL-STD-889	Dissimilar Metals
MS18099	Indicator, Tachometer, Triple Pointer, Helicopter
MS24367	Lamp, Incandescent-Miniature Integral Lighting
MS24515	Lamp Sub-Miniature
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3.1 and 6.2).

3.2 Selection of Government documents. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties shall be used, where on the date of invitation for bids, there are no suitable standard parts. In any case, commercial parts such as screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

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a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.3.1 Critical materials Non-critical materials shall be used where practicable Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Non-magnetic materials. Non-magnetic materials shall be used for all parts of the tachometer indicator, except where otherwise specified.

3.3.3 Metals. Metals shall be of the corrosive resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions, to which the switch may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals Dissimilar metals as defined in MIL-STD-889, shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings. Protective coatings will not be necessary if the metals are in a sealed atmosphere as described in 3.4.3.2.

3.3.3.2 Magnesium alloy parts. Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.3 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.4 Iron and steel parts. Iron and steel parts shall be chromium, nickel or zinc plated in accordance with QQ-C-320, QQ-N-290 or QQ-Z-325 respectively. The class and type of plating shall meet the requirements of paragraph 3.3.4.

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3.3.4 Protective treatment. When materials used in the construction of the indicator are subjected to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

3.3.5 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.6 Fumes and vapors. Materials used in the construction of the indicator shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.4 Design and construction. The indicator shall be designed for operation from the varying frequency alternating current output of tachometer generators conforming to MIL-G-9398 or MIL-G-26611. The frequency of the generator output shall be equal to the speed at which the generator is rotating. The design shall be such that with the increasing and decreasing speeds the pointers shall move smoothly at a rate consistent with the speed change. For the respective models, 100 percent RPM shall be as classified in 1.2, the indicator shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations and such other conditions as are incident to service, shipping, storage, and installation without failure.

3.4.1 Maintenance The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.2 Stops. Each mechanism of the indicator shall be provided with a stop which upon reverse operation will stop the pointer motion between -2 and -5 percent indication.

3.4.3 Case. The outline dimensions of the case shall conform to MS18099. The case of the indicator shall be made of non-ferrous metal, uniform in texture, having a smooth surface and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.4.3.1 Hermetic sealing. The case shall be hermetically sealed and the design shall be such that the mechanism may be removed from the case, replaced, and the case resealed at least three times. The hermetic seal shall be a perfectly closed and airtight seal made

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between metallic and metallic, or metallic and vitreous materials. A hermetic seal is not intended to include seals accomplished by gaskets. The removal of the mechanism and resealing of the case shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity.

3.4.3.2 Filling medium. The filling medium shall be a mixture of 90 percent nitrogen and 10 percent helium or a filling medium approved by the procuring activity. The nitrogen used shall be in accordance with BB-N-411 Type I, Class 1, grade C. The filling medium shall contain not more than .006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ± 0.1 atmosphere.

3.4.4 Weight. The indicator shall be so designed that when completely assembled the weight shall not exceed 3.0 pounds.

3.5 Electrical requirements.

3.5.1 Integral lighting. The MS180999-2 tachometer indicator shall be provided with integral lighting in accordance with MIL-L-25467.

3.5.2 Lamps. The lamps shall be located within the indicator case. Lamps conforming to MS24367 or MS24515-1 may be used in lieu of, or in addition to the lamps specified in MIL-L-25467.

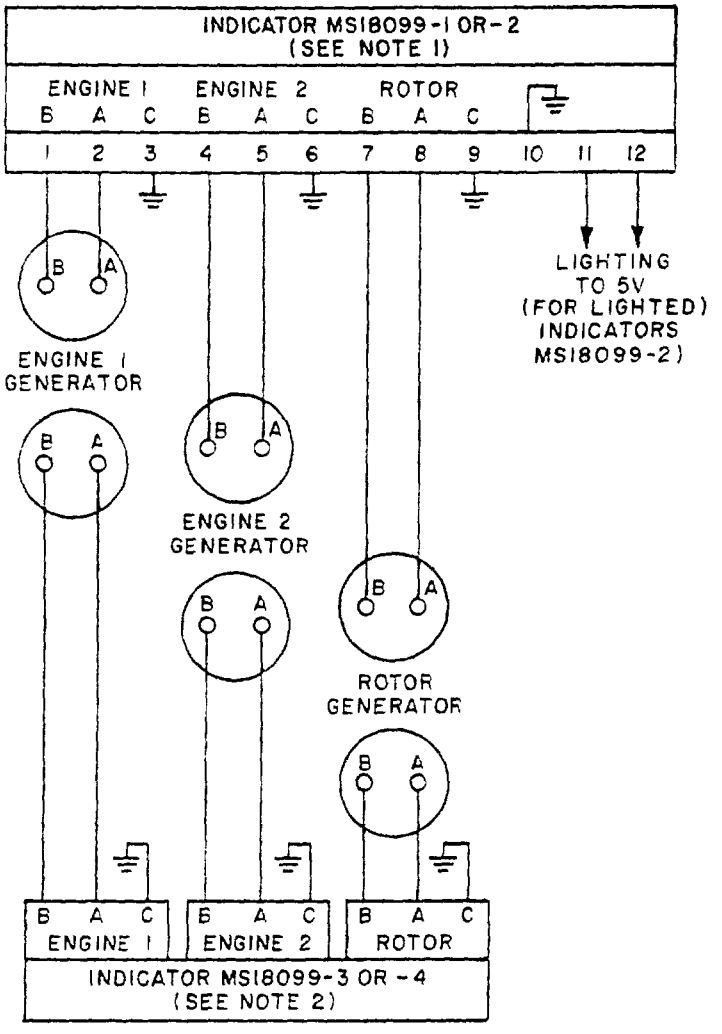
3.5.3 Electric connectors. The electric connector used shall conform to MIL-C-26482 and MS18099.

3.5.4 Wiring. The external wiring shall be as shown in Figure 1. Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable. The wiring shall be insulated from the instrument case. The three leads from the indicator windings shall be designated as B, A, and C (conforming to phases 1, 2, and 3). The leads shall be connected to pins of the electric connector in such a manner that the indicator reading is positive when 3-phase power with phase sequence B-A-C is applied. The wiring shown in Figure 1 is for counterclockwise generator rotation (when viewed from the shaft end). For clockwise rotation it is necessary to reverse leads A and B at the generator plug.

3.5.5 Bearings. The ball bearings used in the indicators shall be ABEC-5P, or better, and shall comply with the requirements of MIL-B-81793.

3.5.5.1 Bearing lubricant. Unless selected from the lubricants listed in MIL-B-81793, the bearing lubricant shall be subject to approval by the procuring activity. Sufficient engineering and test data shall be supplied to demonstrate the adequacy of the lubricant for which approval is requested. If a light instrument oil or silicone oil is

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Note 1. Two MS-18099-1 and/or MS18099-2 indicators may be connected in parallel. DO NOT PARALLEL WITH THE -3 OR -4.

Note 2. The MS18099-3 indicators may be connected in parallel. The MS-18099-4 indicators may be connected in parallel. DO NOT PARALLEL THE MS18099-3 AND -4; DO NOT PARALLEL WITH MS18099-1 OR -2.

THE WIRING DIAGRAM SHOWS THE CONNECTIONS WHEN THE GENERATOR DRIVE SHAFTS TURN COUNTERCLOCKWISE (WHEN VIEWED FROM THE SHAFT END). THE GENERATOR SHELLS ARE THE "C" CONNECTIONS AND SHOULD BE GROUNDED TO COMPLETE THE "C" CIRCUITS.

FIGURE 1. WIRING

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used, barrier coating in accordance with MIL-B-81793 shall be applied to the bearings. Bearings which are not barrier coated shall be cleaned and relubricated just prior to installation in the indicator.

3.6 Dial and pointer. The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used they shall be so located that they will not interfere with the proper location of any dial marking.

3.6.1 Visibility of dial. The pointers, numerals, at least 1/3 of the shortest graduations and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.6.2 Dial marking. The dial shall be marked as shown in Figure 2. Unless otherwise specified, all visible portions of the dial shall be lusterless black Color No. 37038 of FED-STD-595 and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. All markings shall be durable to withstand usage encountered in service. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which it applies. If practicable, each numeral shall be placed so that the center of area of the numeral is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64th inch.

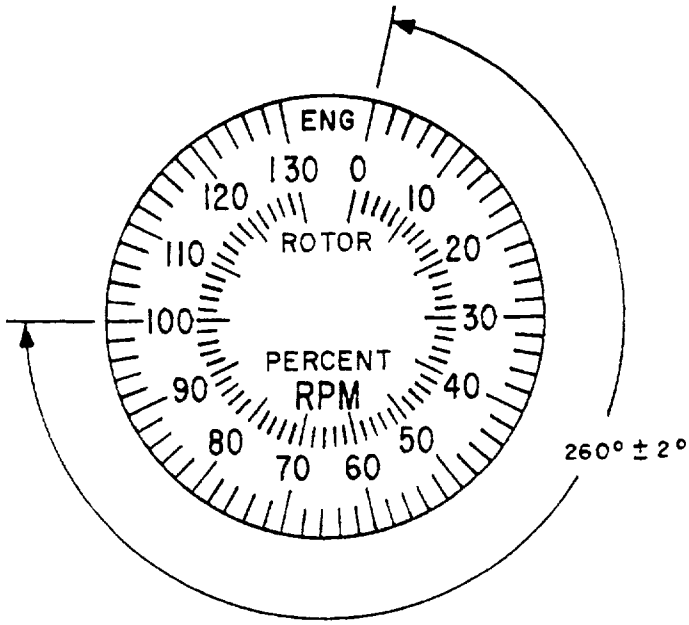
3.6.3 Pointers. The pointers of the indicator shall be as light weight as practicable, sufficiently rigid to prevent oscillation under vibration and shall be firmly attached to the mechanism but shall also be readily adjustable. The pointers shall conform to the requirements of MS33585-6, marked with an "R", for the front pointer, MS33585-12 for the center pointer, and MS33585-11 for the rear pointer. The finish on the shaded portion of the pointers shall be lusterless white conforming to Color No. 37875 of FED-STD-595. Pointer MS33585-12 shall have a black longitudinal line 0.012 inch thick running through the center of the pointer between the marking "2" and the tip. For the MS18099-2 indicators the pointers shall be suitable for integral lighting.

3.7 Cover glass. The cover glass shall be in accordance with MIL-G-81704, Type III. Any flaws permitted in MIL-G-81704 shall not interfere with reading the dial.

3.7.1 Cover glass mounting. The distance from the inner surface of the cover glass to the surface of the dial on which the marking is applied shall not exceed 0.250 inch.

3.7.2 Reflection reducing coating. The cover glass shall be coated on all glass-to-air surfaces with multiple-layer, low-reflection coating conforming to MIL-C-14806.

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	HEIGHT, LENGTH OR DIAMETER ±.010*	WIDTH OF LINE OR GRADUATION ±.005	FINISH
ENGINE SCALE O.D.	2.760 +.000 −.110	----	COLOR 37875, LUSTERLESS WHITE Fed. Std. No. 595
MAJOR GRADUATIONS	.218	.032	
MINOR GRADUATIONS	.160	.015	
NUMERALS	.187	----	
ROTOR SCALE O.D.	1.688 +.000 −.068	----	
MAJOR GRADUATIONS	.187	.032	
MINOR GRADUATIONS	.125	.015	
LETTERING			
RPM	.187	----	
PERCENT, ENG, ROTOR	.125	----	

*UNLESS OTHERWISE SPECIFIED

FIGURE 2. DIAL MARKINGS

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* 3.8 Performance.

* 3.8.1 Dielectric strength. The dielectric strength shall not breakdown or show any other permanent damage to the indicator when tested as specified. (See 4.6.1)

* 3.8.2 Examination. When examined as specified the indicator shall meet the requirements for materials, design and construction, marking and workmanship. (See 3.1, 3.4, 3.9, 4.3, 4.6.2)

3.8.3 Starting. The synchronous speed of the indicator shall not be in excess of 350 RPM when operated at ambient room temperature. (See 4.6.3)

3.8.4 Scale error The indicator shall be operated at the points of the scale indicated in Table I for room temperature. (See 4.6 4)

3.8.5 Friction error. The indicator shall not have a friction error in excess of 0.9 percent at the 20 percent graduation and 0.7 percent at the 40, 80, 90, 100 and 110 percent graduations. (See 4.6 5)

3.8.6 Position error. The maximum deviation from the normal position reading shall not exceed 0.3 percent when the indicator is tested as specified. (See 4.6.6)

3.8.7 Acceleration. Each movement shall attain synchronism within 2.5 seconds, and the indication shall be in accordance with the room temperature indication as given in Table I for the RPM actually used. (See 4.6.7)

3.8.8 Integral lighting. Each lighted indicator shall be tested in accordance with MIL-L-25467. (See 4.6.8)

3.8.9 Case leakage The rate of leakage of the gas or mixture of gases in the case of the indicator, adjusted to a pressure differential of one atmosphere, shall not exceed one micron cubic foot per hour (10^{-5} cubic centimeters per second) (See 4.6.9)

3.8.10 Magnetic effect. When tested as specified the indicator shall not cause more than a 2 degree deflection of a magnetic compass. (See 4.6.10)

3.8.11 Precipitation or fog. When tested as specified the indicator shall show no evidence of moisture or other deposits on the inner face of the cover glass. (See 4.6.11)

3.8.12 Vibration resonance. When vibrated as specified any readings shall not vary more than 0.7 percent from normal. The amplitude of the pointer oscillation shall not exceed 0.7 percent. (See 4.6.12)

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TABLE I. CALIBRATION SCALE ERRORS

GENERATOR DRIVE SHAFT SPEED (RPM)					CORRECT INDICATED SPEED	TOLERANCE (± Percent RPM)		
MS18099-1, -2	MS18099-3		MS18099-4					
Engine 1 and 2 and Rotor	Engine 1 and 2	Rotor	Engine 1 and 2	Rotor	Percent RPM	RT	-5°C	-54°C +71 C
0	0	0	0	0	0	0.6	0.6	0.6
420	379.4	390	390		10	0.6	0.9	1.6
630	569.1	585	585	325	15	0.8	-	-
840	758.8	760	780	528	20	0.8	-	-
1260	1138.2	1170	1170	792	30	0.8	-	-
1680	1517.6	1560	1560	1056	40	0.8	1.0	1.6
2100	1807.0	1950	1950	1320	50	0.8	1.0	1.6
2520	2276.4	2340	2340	1580	60	0.8	-	-
2840	2655.8	2730	2730	1848	70	0.8	-	-
3150	2845.5	2925	2925	1980	75	0.6	0.9	1.4
3360	3035.2	3120	3120	2112	80	0.6	0.9	1.4
3570	3224.9	3315	3315	2244	85	0.6	0.9	1.4
3780	3414.6	3510	3510	2376	90	0.6	0.9	1.4
3990	3604.3	3705	3705	2508	95	0.5	0.8	1.3
4200	3794.0	3900	3900	2640	100	0.5	0.8	1.3
4410	3983.7	4095	4095	2772	105	0.5	0.8	1.3
4620	4173.4	4290	4290	2904	110	0.7	-	-
4830	4363.1	4485	4485	3036	115	1.0	-	-
5040	4552.8	4680	4680	3168	120	1.0	1.4	1.8
5250	4742.5	4875	4875	3300	125	1.0	-	-
5460	4932.2	5070	5070	3432	130	1.0	1.4	1.8

The above table shows generator drive shaft speeds and the respective equivalent correct indicated percent RPM. As an alternate, the drive shaft speed may be set at hundreds of RPM nearest the respective value shown, the correct indicated percent RPM shall then be calculated, and the respective tolerances shown above shall apply.

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3.8.13 Current consumption. When tested as specified the current consumption shall not exceed 0.30 ampere. (See 4.6.13)

3.8.14 Indicator temperature rise. When tested as specified the temperature rise shall not exceed 60°C. (See 4.6.14)

3.8.15 Scale error at -5°C When tested as specified the scale error at -5°C shall not exceed that specified in Table I at -5°C. (See 4.6.15)

3.8.16 Starting-low temperature. The indicator shall be tested as specified when at -54°C ± 2°C (See 4.6.16)

3.8.17 Scale error at -54°C. When tested as specified the indicator error shall not exceed the tolerances specified in Table I for -50°C. (See 4.6.17)

3.8.18 Scale error at high temperature. When tested as specified the readings at +71°C temperature shall not exceed the tolerances specified in Table I for +71°C. (See 4.6.18)

3.8.19 Low temperature exposure The indicator when tested as specified shall not be adversely affected by low temperature. (See 4.6.20)

3.8.20 High temperature exposure The indicator when tested as specified shall not be adversely affected by high temperatures. (See 4.6.20)

3.8.21 Thermal shock. The indicator when subjected to the tests specified (4.6) shall display no evidence of moisture penetration or damage to finish or enclosures. (See 4.6.21)

3.8.22 Vibration cycling. The indicator when tested as specified shall have a room temperature scale test conducted. (See 4.6.22)

3.8.23 Overspeed and reverse operation. When tested as specified, any change in speed at the test points before and after the overspeed and reverse operation test shall not be more than 0.5 percent. (See 4.6.23)

3.8.24 Lamp life. The indicator shall meet the requirements specified in MIL-L-25467.

3.8.25 Life test. The indicator when tested as specified shall be capable of operating for 1000 hours at an indicated speed of 90 ± 5 percent without any detrimental affects to its operation. (See 4.6.25)

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3.8.26 Fungus resistance. The indicator when tested as specified shall not show evidence of the growth of fungi. (See 4.6.26)

3.8.27 Salt spray. The indicator when tested as specified shall not show any evidence of external corrosion or deterioration. (See 4.6.27)

3.8.28 Reliability. Indicator reliability shall be analyzed, calculated, and reported in accordance with MIL-STD-756.

3.8.28.1 Reliability in mean-time-between-failures. The indicator lower test mean-time-between-failures (61 MTBF), as defined in MIL-STD-781, shall be 1000 hours for testing and acceptance as specified in 4.4.3.

3.9 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacture's part number shall be governed by the drawing number requirements of MIL-D-1000.

3.10 Identification of product. A nameplate shall be securely attached to the exterior of the case and shall be legibly and durably marked in accordance with MIL-STD-130 and shall include the information specified in the applicable specification sheet.

3.11 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished to produce an item free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.11.1 Soldering. Soldering shall be performed in accordance with MIL-S-6872.

3.12 Screw threads. Screw threads 0.060 inch in diameter or larger shall be in accordance with specification MIL-S-7742.

3.13 No data is required by this specification (other than reports accompanying samples submitted for qualification testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see paragraph 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the indicator shall be classified as follows:

a. First article inspection (4.3)
b. Quality conformance inspection (4.4)

4.3 First article inspection. First article inspection shall consist of all the examinations and tests of this specification.

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after award of a contract or order, the manufacturer shall submit three indicators of each part number for first article testing. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these indicators from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article samples shall be furnished to the Government as directed by the contracting officer (see 6.2). Each sample size shall be plainly identified by a securely attached durable tag marked with the following information:

Sample for first article
Submitted by (name) (date) for
tests in accordance with the
requirements of specification
MIL-I-23832 (reference letter
authorizing test)
Indicator, Tachometer, Triple Pointer,
Helicopter
Manufacturer's Part No.
MS Part No.
Name of manufacturer

4.3.1.1 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the Government quality assurance representative. The samples may be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract (see 6.2).

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4.4 Quality conformance inspection. The quality conformance inspections of the indicator shall consist of the individual tests and the sampling plans of this specification.

4.4.1 Individual inspection. Each indicator submitted for acceptance shall be subjected to the Individual inspection. This inspection shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each indicator accepted shall have passed the following inspections:

Dielectric strength
Examination of product
Starting - room temperature
Scale error - room temperature
Friction error
Position error
Acceleration
Integral lighting (for lighted indicators only)
Case Leakage

4.4.2 Sampling plans. The Sampling plans shall consist of Sampling plan A and Sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual inspections. The inspection samples which have been subjected to Sampling plan A inspection shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling plan B inspection shall not be delivered on contract.

4.4.2.1 Sampling plan A sample selection. Sampling plan A samples shall be selected at random in accordance with the following schedule

Quantity Offered for Acceptance	Quantity to be Selected for Inspection
First 15	(See Note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

NOTE: When Sampling plan B is invoked, the quantity shall be zero. When Sampling plan B is to be omitted the quantity shall be one.

When a defective indicator occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of fifteen (when Sampling plan B is omitted) and proceed as indicated.

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4.4.2.1.1 Sampling plan A inspection. Each sample selected for Sampling plan A inspection shall be subjected to the following tests:

Precipitation or fog
Magnetic effect
Vibration resonance
Current consumption
Indicator temperature rise
Scale error at -5°C
Starting - low temperature
Scale error at -54°C
Scale error - high temperature

4.4.2.2 Sampling plan B instructions. Two indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture These samples shall be forwarded at the contractor's expense to a Government Laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information.

INDICATOR, TACHOMETER, TRIPLE POINTER, HELICOPTER
Submitted by (Manufacturer's name, date) for production acceptance Sampling plan B testing, in accordance with Contract No. _____
Manufacturer's part number _____

4.4.2.2.1 Sampling plan B inspection. Each indicator selected for Sampling plan B testing shall be subjected to the following tests:

Sampling plan A
Low temperature exposure
High temperature exposure
Thermal shock
Vibration cycling
Overspeed and reverse operation
Lamp life (for lighted indicators only)
Life
Test for helium
Internal examination
Lighting contrast (for lighted indicators only)

4.4.2.3 Individual and sampling plan test procedures. A detailed test procedure which supplements the inspection methods of 4.6 shall be prepared for the type of inspection to be conducted prior to starting tests. The contractor shall be responsible for preparing the test procedures for all inspections not conducted at a Government laboratory. The procedures shall be submitted to and approved by the procuring activity prior to use. The procedure shall state exact conditions under which measurements are to be made and shall evaluate all requirements and performance characteristics. The procedures shall contain all necessary data sheets needed to record the conditions and measurements.

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4.4.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781 and shall consist of the first article (preproduction) reliability qualification test and the all equipment production reliability test. Indicators selected for the reliability assurance tests shall first have passed the individual tests. Reliability prediction, a reliability test plan document, and a burn-in test are not required. The thermal survey report and vibration survey report may be incorporated as appendices to the reliability test procedures. A test procedure shall be prepared for both the first article and all equipment tests.

4.4.3.1 First article (preproduction) reliability qualification test. A minimum of three and a maximum of six indicators shall be tested. The indicators selected for this test shall be from the first 15 produced on the contract or order. The indicators shall be tested as outlined in MIL-STD-781. For both the first article and all equipment reliability tests, the temperature shall be cycled between -54°C and +71°C, which shall be time to stabilize at low temperature followed by time to stabilize at high temperature plus two hours. Vibration shall be 2.2g ± 10 percent peak acceleration at any nonresonant frequency between 20 and 60 Hz, measured at the mounting points of the instrument. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The indicators shall be off during the cooling portion and operating during the heating portion of the temperature cycle. For lighted indicators, the input lighting power shall be 4.5 ± 0.5V dc, or 400 Hz ± 5 Hz ac. The accept/reject criteria of Test Plan IIIC shall be used to determine the length of the test.

4.4.3.2 All equipment production reliability test. The indicators shall be tested as specified in MIL-STD-781. The test conditions specified in 4.4.3.1 shall apply. Test Plan XVIIIC shall be used.

4.4.3.2.1 Procedure for the all equipment reliability test. This test shall be conducted on all indicators submitted for acceptance on the contract or order except those submitted for the first article (preproduction) reliability qualification test. It may be conducted concurrently with the reliability qualification test. Each indicator shall be tested for 72 hours ("on time"). If desired, a burn-in period may be included prior to this test. However, if the burn-in period is used, it must be uniform in time and severity for each indicator. To determine if the MTBF is being met at any time during the contract, the operation test hours ("on time") and the failures (not counting burn-in operation time or failures) shall be totaled and the results compared with the reject line of Test Plan XVIIIC of MIL-STD-781 (extend the accept and reject lines as necessary to accommodate the data). These totals shall be accumulative so that the experience from the beginning of the contract is always included. At the end of each month, the results shall be sent to the procuring activity. At any time the current

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totals of test hours and test failures plotted on Test Plan XVIIIC curves show a reject situation, the procuring activity shall immediately be notified. The procuring activity reserves the right to stop acceptance of indicators at any time a reject situation exists pending review of the supplier's efforts to improve the equipment, equipment parts, workmanship, etc., so that the entire compilation will show other than a reject decision

4.4.3.3 Test procedures and test details. Test procedures to be used for each phase of the reliability assurance tests shall be prepared for the supplier. Each procedure shall be submitted to and approved by the procuring activity before that phase of the reliability assurance testing contained in the procedure may begin. The test procedures shall contain all pertinent test details such as the length of the duty cycle, length of the heating and cooling portions of the cycle, performance characteristics to be measured, special failure criteria, burn-in period if used, test equipment and wiring diagrams to be used, etc. The following paragraphs shall be considered as minimum requirements and shall apply to both phases.

4.4.3.3.1 Duty cycle. The duty cycle shall consist of supplying the indicator with signal inputs from separate generators conforming to MIL-G-9398 or MIL-G-26611, or equivalent power source during the heating portion of the temperature cycle at an indication of 90 $\pm$ 5 percent RPM. The lighted indicator lighting shall be energized when the indicator is operating.

4.4.3.3.2 Performance characteristics. The scale error and friction error tests of 4.6.4 and 4.6.5 shall be conducted at least once each week on the indicators being subjected to the reliability qualification test and at the end of the 72 hours on the indicators being subjected to the reliability all equipment test. These scale and friction errors shall not exceed those specified herein for room temperature. In addition, once each working day (a minimum of once for reliability production indicators) the scale error of each indicator shall be determined at the 90 percent RPM test point within five minutes after the duty cycle begins and again at the end of the duty cycle for each movement. The -54°C/71°C tolerance shall apply. The indicators shall also be observed for pointer oscillation. Pointer oscillation shall at no time exceed the tolerance specified in 4.6.4.1.

4.4.3.3.3 Failure criteria. Whenever performance characteristics fall below the requirements of 4.4.3.3.2 above, at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless the procuring activity and the supplier agree that one part caused the other parts to fail.

4.4.3.3.4 Preventative maintenance. No preventative maintenance may be accomplished on the indicators while they are on test.

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4.4.3.4 Disposition of indicators upon completion of tests. Any indicator used for the reliability assurance test may be delivered on contract provided it meets all of the following requirements:

a. It is representative of production units currently being accepted.

b. It is in "good as new" condition or has been refurbished (see 6.3).

c. It is otherwise satisfactory.

4.5 Inspection conditions.

4.5.1 Standard conditions. Unless otherwise specified, all inspections required by this specification shall be made under the following conditions:

Temperature	Room ambient, $25^{\circ}C \pm 5^{\circ}C$
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90% relative humidity

4.5.2 Test readings. Unless otherwise specified, before a test reading is taken, the indicator shall be vibrated using an electric vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.5.3 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.5.4 Electrical power. Unless otherwise specified, tests requiring the operation of the indicator shall be conducted utilizing the output of tachometer generators conforming to MIL-G-9398 or MIL-G-26611. The output frequency shall be monitored to assure that it is within ± 0.1 percent, based on 100 percent RPM, at any test point.

4.5.5 Pointer or movement interference. In tests that require operation of the indicator, the inspection methods are intended for each movement of each indicator. Tests of the movements may be conducted individually or simultaneously unless otherwise specified; however, a check of each movement through its entire range to assure that there is no interaction with any other movement or other interference shall be conducted in each operational test.

4.5.6 Integral lighting. The lighted indicator lighting system shall be energized and shall operate satisfactorily during all tests except fungus resistance, salt spray and as otherwise specified herein. The lighting shall operate satisfactorily after completion of each test unless otherwise specified herein.

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4.6 Inspection methods.

4.6.1 Dielectric strength. A potential of 200 volts alternating current at commercial frequency shall be applied between isolated pins and between pins and case for a period of 5 seconds. (See 3.8.1)

4.6.2 Examination of product. Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests. (See 3.8.2)

4.6.3 Starting. The indicator when at ambient room temperature shall be connected to a variable frequency power supply. The supply voltage shall be 0.0035 times the frequency in hertz. The frequency of the supply shall be slowly increased from zero hertz and the speed at which the indicator begins to operate at synchronous speed shall be noted.

4.6.4 Scale error. The indicator, at ambient room temperature shall be connected as shown in Figure 1 to a generator conforming to MIL-G-9398 or MIL-G-26611 and tested at points of the scale indicated in Table I for room temperature. The indicator shall have been at room temperature for at least four hours prior to test. The test shall be made either by driving the generator at speeds necessary to produce the specified indicator readings and reading the speed of the generator, or by driving the generator at the specified speeds and reading the indicator. The test shall be made with speeds increasing and with speeds decreasing. With speeds increasing, the speed shall be brought up to, but shall not exceed, the speed specified to obtain the desired reading; with speeds decreasing, the speed shall be brought down to but shall not fall below, the speed specified to give the desired reading.

4.6.4.1 The room temperature scale errors at the test points shall not exceed the tolerance specified in Table I. When the speed is held constant at any test point, the pointer shall not oscillate over a range greater than 0.5 percent from 0 to 15 percent reading and 0.3 percent from 15 to 120 percent reading. (See 3.8.4)

4.6.4.2 In performing the tests specified for the scale error test, two movements of the indicator shall be connected to, and operated from, one tachometer generator.

4.6.5 Friction error. The indicator shall be set at approximately the 20, 40, 80, 90, 100, and 110 percent graduations. After each setting, the indicator shall be vibrated as specified in 4.5.2 and change in indication noted. (See 3.8.5)

4.6.6 Position error. The indicator shall be operated at 100 percent in the normal (dial vertical) position. The indicator reading shall be noted. The indicator shall be tested in the following positions:

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a. Rotated from normal position through 45 degrees to the rear (dial tilted 45 degrees back from vertical).

b. Rotated from normal position through 45 degrees to the front (dial tilted 45 degrees forwarded from vertical).

c. Rotated from normal position through 45 degrees to the right (dial vertical).

d Rotated from normal position through 45 degrees to the left (dial vertical). (See 3.8.6)

4.6.7 Acceleration. A generator connected to the movement undergoing test, shall be brought up to the equivalent of approximately 90 percent RPM (in accordance with Table I) in 1/2 second. The generator voltage shall have been adjusted so that it is in the same proportion at the actual RPM used as 21 ± 0.5 volts is to 4200 RPM, when connected to the movement under test. (See 3.8.7)

4.6.8 Integral lighting. Each lighted indicator shall be subjected to and shall meet the requirements of the following tests specified in MIL-L-25467.

(1) Operation
(2) Dial visibility

4.6.9 Case leakage. The indicator shall be tested for case leakage with a mass spectrometer or an equivalent, at least equally sensitive, quantitive leakage test.

4.6.10 Magnetic effect. The indicator shall be properly connected and power applied (100% indication). The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270 and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. This test shall be repeated with no power applied and the same tolerance shall apply. (See 3.8.10)

4.6.11 Precipitation or fog The indicator shall be placed in a test chamber. The temperature of the circulating air shall be 85° ± 5°C and the indicator shall be kept in the chamber for 2 hours. The indicator shall then be removed from the chamber and within 30 seconds the indicator face shall be immersed to a depth of 1 to 2 inches in a fluid bath maintained at 0° ± 5°C. After 5 minutes, the indicator shall be removed from the bath and dried. (See 3.8.11)

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4.6.12 Vibration resonance. The indicator reading shall be determined when the generator is operated at a point between 60 and 90 percent RPM. The indicator shall then be tested in accordance with the resonance portion of Procedure XIII of MIL-E-5272 and the indicator shall be monitored while the generator is operating at the original speed. (See 3.8.12)

4.6.13 Current consumption. The MS18099-1 or -2 indicator shall be connected to a 21 volt, 4200 hertz power supply. The current drawn by any one of the 3-phase lines shall not exceed 0.30 ampere. The MS18099-3 or -4 indicator input shall be at 100 percent RPM from separate generators for each movement. (See 3.8.13)

4.6.14 Indicator temperature rise. The indicator shall be placed in a normal dial-vertical position where there is no circulation of the air surrounding the indicator, with the exception of thermal current caused by the heat dissipated from the indicator. All three movements of the MS18099-1 or -2 indicator shall be operated simultaneously at 4200 $\pm$ 5 RPM at a voltage of 21 $\pm$ 0.1 volt for a period of one hour, after which the temperature rise shall not exceed the temperature specified. The MS18099-3 or -4 indicator shall be operated at 100 percent RPM from separate generators for each movement. (See 3.8.14)

4.6.14.1 The temperature rise of the indicator shall be determined by change in resistance of the copper windings. The temperature rise (T-t) is calculated by the following equation:

$$T-t = \left(\frac{R}{r} - 1\right)(234.5 + t)$$

Where:

t = the ambient temperature and the temperature of the stator winding at the start of the test in degrees centigrade.

T = the final temperature of the stator windings, degrees centigrade.

r = the direct current resistance of the stator windings at temperature t, measured between 2 terminals.

R = the direct current resistance at temperature T measured between the same 2 terminals.

4.6.15 Scale error at -5°C. The indicator readings shall be determined as specified in 4.6.3 except the indicator shall be at -5° $\pm$ 2°C. This temperature shall be maintained at least 2 hours before readings are taken. (See 3.8.15)

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4.6.16 Starting-low temperature. Prior to this test, the scale error test at room temperature shall be repeated. The indicator shall then be placed in a test chamber and the temperature of the circulating air shall be reduced to $-54^{\circ} \pm 2^{\circ}C$. The indicator shall be kept in the chamber for 4 hours after the test temperature is reached. No voltage shall be applied to the indicator terminals during this period. (See 3.8.16)

4.6.16.1 After 4 hours at the test temperature, one movement of the indicator shall be connected to a variable frequency power supply. The voltage of the supply shall be equal to 0.004 times the frequency in hertz and the speed at which the indicator begins to operate at synchronous speed shall be noted. This speed shall not exceed 350 RPM. Each of the three movements shall be tested in its turn. (See 3.8.16)

4.6.17 Scale error at $-54^{\circ}C$. Following the starting test at low temperature and while still at $-54^{\circ}C$, the indicator readings shall be determined as specified in 4.6.3. Upon return to room temperature the indicator shall be subjected to, and meet the requirements of the scale error-room temperature test. (See 3.8.17)

4.6.18 Scale error-high temperature. The indicator readings shall be determined as specified in 4.6.3 except that the indicator shall be tested at $71^{\circ} \pm 2^{\circ}C$ and shall be at that temperature at least 4 hours before readings are taken. Upon return to room temperature, the indicator shall be subjected to, and meet, the requirements of starting low temperature (4.6.16) and scale error at $-54^{\circ}C$ (4.6.17). (See 3.6.18)

4.6.19 Low temperature exposure. The indicator shall be subjected to a temperature of $-62^{\circ} \pm 2^{\circ}C$ for a period of 48 hours. The temperature shall then be raised to $-54^{\circ} \pm 2^{\circ}C$ for a period of 24 hours. At the end of this period, and while still at this temperature, the indicator shall be subjected to, and meet, the requirements of low temperature, and scale error at $-54^{\circ}C$ test. After the chamber has returned to room temperature and the indicator has remained at room temperature for a period of 4 hours it shall be subjected to and meet the requirements specified in 4.6.3, 4.6.4 and 4.6.5. There shall be no damage which would adversely affect subsequent operation. (See 3.8.19)

4.6.20 High temperature exposure. The indicator shall be placed in a test chamber and the temperature of the circulating air shall be increased to $71^{\circ} \pm 2^{\circ}C$ with an internal relative humidity of not more than 5 percent. The source of heat for the chamber shall be arranged in such a manner that substantial radiant heat shall not fall upon the test specimen. The total volume of a single item of equipment or of several items shall not exceed 50 percent of the internal volume of the test chamber. The indicator shall be maintained at $71^{\circ} \pm 2^{\circ}C$ for a period of 24 hours. While still at this temperature, the three movements of the indicator shall be operated simultaneously at 100 percent RPM for one hour, then throughout its range without sticking or

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erratic operation. After the chamber has returned to room temperature and the indicator has remained at room temperature for at least 4 hours, it shall be subjected to and meet the requirements of the Scale error-room temperature, the Friction error and Leakage tests. (See 3.8.20)

4.6.21 Thermal shock. The indicator shall be subjected to 4 cycles of the test. Each cycle of the test shall consist of immersing the indicator in water at 85°C for a period of 30 minutes and then, within 15 seconds of removal from the bath, the indicator shall be immersed in the other bath maintained at 5°C. This cycle shall be repeated continuously, one cycle following the other, until 4 cycles have been completed. After 5 minutes immersion in the 5°C bath during the final cycle, remove the indicator and check for evidence of precipitation or fog inside the cover glass. There shall be no evidence of precipitation or fog. Following the test, the indicator shall be subjected to and shall meet the requirements of the Leakage test (4.6.9). (See 3.6.21)

4.6.22 Vibration cycling. The generator shall then be operated at a speed equivalent to approximately 90 percent RPM, while the indicator is being cycled in accordance with Procedure XIII of MIL-E-5272. Upon completion the indicator shall be subjected to the requirements of the Scale error-room temperature test. (See 3.8.22)

4.6.23 Overspeed and reverse operation. The indicator shall be electrically connected to a tachometer generator and the scale errors of the indicator shall be determined when operated at approximately 20 and 90 percent RPM. The indicator shall be at room temperature and the readings shall be taken before the indicator temperature has risen appreciably.

4.6.23.1 The generator and indicator shall then be operated, first at a generator shaft speed equivalent to approximately 125 percent RPM for a period of 5 minutes, then in the reverse direction of rotation at a generator shaft speed equivalent to approximately 60 percent RPM for a period of one minute.

4.6.23.2 After the indicator has been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at approximately 20 and 90 percent RPM. (See 3.8.23)

4.6.24 Lamp life. The lighted indicator shall meet the Life test requirements specified in MIL-L-25467. This test shall be run concurrently with the Life test. (See 3.8.24)

4.6.25 Life test. The readings of the indicator shall be determined when the generator is operated at 20 and 90 percent RPM. The indicator shall be at room temperature and the readings shall be taken before the indicator temperature has risen appreciably. The three indicator movements shall then be operated simultaneously from separate generators, or equivalent, at an indicated speed of 90 ± 5 percent RPM for a period of 1000 hours. At the end of the 1000 hour operating

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period and after the indicator has cooled to room temperature, the readings of the indicator shall again be determined when the generator is operated at indicated speeds of 20 and 90 percent RPM. The change in readings before and after this test, shall not exceed 0.5 percent. No screws or other parts shall become loose of damaged as a result of this test. Indicators which have been subjected to all the tests specified shall again be subjected to the requirements of the Scale error-room temperature, Friction error and Position error tests. Scale error shall not be 0.5 percent more than allowed at room temperature on Table I. Friction error shall not exceed 1.5 percent at the 0 to 20 percent reading and 1.0 percent at any of the other graduations. The indicator shall be subjected to and pass the requirements of the Position error test, except that the allowed tolerance shall be 0.5 percent. (See 3.8.25)

4.6.26 Fungus resistance. The indicator shall be tested as specified in MIL-E-5272, Procedure I, except that the test period shall be 14 days At the option of the qualifying agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument. At the end of the 14 day test period, the indicator shall be examined for fungi, any evidence of fungi growth shall be cause for rejection. (See 3.8.26)

4.6.27 Salt spray. The indicator shall be subjected to a salt spray test in accordance with Procedure I of Specification MIL-E-5272. External connections shall be made to the indicator in such a manner as to simulate installed conditions. Following this test, the indicator shall be examined and there shall be no evidence of external corrosion or deterioration which would affect subsequent operation. The indicator shall then be subjected to and shall pass the Dielectric strength and Leakage tests as specified herein. The external surface of the indicator may be washed with distilled water and air dried prior to conducting the Dielectric strength test.

4.6.28 Test for helium. The indicator case shall be punctured, or the filling tube cut, and the indicator subjected to the test for helium by means of a mass spectrometer type helium detector. Failure to detect helium shall be cause of rejection.

4.6.29 Internal examination. The case of the indicator shall be opened and the internal mechanism shall be examined. The lamp circuit of the lighted indicator shall also be examined as specified in Specification MIL-L-25467. Any deterioration or damage which could in any manner prevent the indicator from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.6.30 Lighting contrast. The lighted indicator shall be subjected to and meet the requirements of the Contrast test specified in Specification MIL-L-25467, except that the C index number must be at least 70 percent of the minimum respective values.

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5. PACKAGING

5.1 Packaging. The indicator shall be packaged in accordance with MIL-STD-794, Level "A". The method of preservation shall be in accordance with MIL-P-116, Method 1A-5 without preservative compound, using metal reusable containers.

5.2 Packing. The indicator shall be packed in accordance with MIL-STD-794, Level A, B or C as specified in contract or order. (See 6.2)

5.3 Marking. The interior and exterior containers shall be marked as specified in MIL-STD-794.

5.3.1 Precautionary marking The following precautionary markings shall appear on two sides of each interior package and shipping container whenever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use. The indicator covered by this specification is intended for use in helicopters to indicate percent RPM.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, Number and Date of this specification.

b. The quantity, and MS part number including dash number of the instrument desired.

c. Level of packing (see 5.2) desired.

d. Data requirements. Certain data is required to implement 3.8.1, 4.4 and 4.4.3.

e. Whether first article inspection is required and where the samples should be delivered (4.3.1 and 4.3.1.1).

f. Whether first article inspection may be waived (see 4.3.1).

6.3 Definitions.

6.3.1 Amplitude. Whenever the word "amplitude" is specified it shall mean the extent of motion as measured from one extreme to the opposite extreme.

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6.3.2 Hermetic seal. A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.3 Movement. For purposes of this specification the indicator is considered to have three movements. These movements operate respectively, the rotor pointer and pointers 1 and 2.

6.3.4 Good as new. "Good as new" shall mean indicators operated less than 10 percent of specified MTBR operation.

6.3.5 Refurbished. "Refurbished" shall mean the indicator has been completely overhauled with all component parts meeting current parts standards; and the indicator shall have been subjected to, and met, all the requirements of a new indicator.

6.4 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirement List (DD Form 1423) incorporated into the contract When the provisions of DAR-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4 4.2.3	Individual and Sampling Test Procedures	UDI-T-21347 - Test Procedures
4.4 3	Thermal Survey Report	DI-R-7036
4 4.3	Vibration Survey Report	DI-R-7037
4.4.3.3	Reliability Test Procedures	DI-R-7035
4.4.3	Reliability Test and Demonstration Reports	DI-R-7034
4.4.3	Plan, Corrective Action	DI-R-7038
4.4.3	Report, Failed Item Analysis	DI-R-7039
4.4.3	Report, Failure Summary and Analysis	DI-R-7041

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6.5 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract. The contract shall have precedence over any specification.

(2) This specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviations from this specification, or from subsidiary specifications where applicable shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications. Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 When a First Article is required, it shall be tested under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The First Article should be a preproduction sample. The contracting officer should include specific instructions for examinations, tests, and approval of the first article.

Custodians:
Navy - AS

Preparing Activity
Navy - AS
(Project No. 6620-N051)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

MIL- I- 23832A (AS)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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